Clustering and Customer Segmentation: Final Project Report

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Project 8: Clustering and Customer Segmentation
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Executive Summary

This culminating report will show the final K-Means clustering of customer segmentation with the Online Retail II data in Altair AI Studio (RapidMiner). Continuing on the midterm progress report, the project team completed the clustering workflow, tested the decision to use K = 5 by scenario analysis against K = 4, and created five business-interpretable segments of the customer population: VIP, Loyal, Regular, Occasional, and Inactive Customers.

The last cluster model identified 5860 customers in five clusters using the normalized Recency, Frequency, and Monetary (RFM) values. The still base is primarily made up of Loyal Customers (52.8% of base), with recurring revenue anchored, and a small, yet distinct, VIP segment (0.2) providing disproportionate value per customer. The following report presents the final workflow, cluster solution, and visualizations, segment-specific advice, business value, and constraints of the analysis.

Introduction

Customer segmentation transforms raw transaction data into useable segments that can be utilized by organizations in order to target marketing, keep high-value customers, and efficiently allocate resources. This project utilized the K-Means clustering of the Online Retail II dataset on the basis of Recency, Frequency, and Monetary (RFM) analysis, which transforms the transaction-level customer records into behavioral profiles at the customer level.

The mid-term of this project defined the data cleaning pipeline, feature engineering in the form of RFM, normalization strategy (Range Transformation), and a preliminary K = 5 clustering solution. This final report substantiates that work: it reports the final workflow as implemented, justifies the number of clusters by comparing K = 4 and K = 5 scenarios, profiles every segment separately, and puts the findings into specific business advice form.

Final Clustering Workflow

The last workflow generated in Altair AI Studio (RapidMiner) includes seven operators that are run sequentially: Read Excel, Aggregate, Rename, Generate Attributes Recency, Select Attributes, normalize (Range Transformation), and K -Means Clustering. The structure of this pipeline remained the same as during the midterm stage; the normalization technique (Range Transformation over Z-Transformation) was used due to the testing that it yielded more balanced and interpretable clusters.

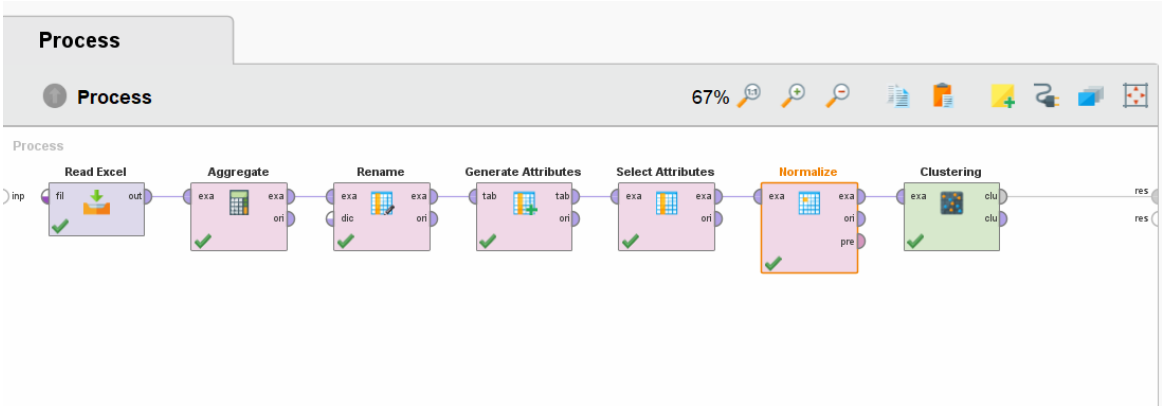


Figure 1. Final RapidMiner workflow: Read Excel → Aggregate → Rename → Generate Attributes → Select Attributes → Normalize → Clustering.

Final Cluster Solution

The normalized RFM variables were clustered using K-Means (K = 5), and created 5 clusters out of 5,860 total customers. The sizes of clusters were varied, with one small and high-value cluster comprising 10 customers, and the large dominant cluster of 3,096 customers.

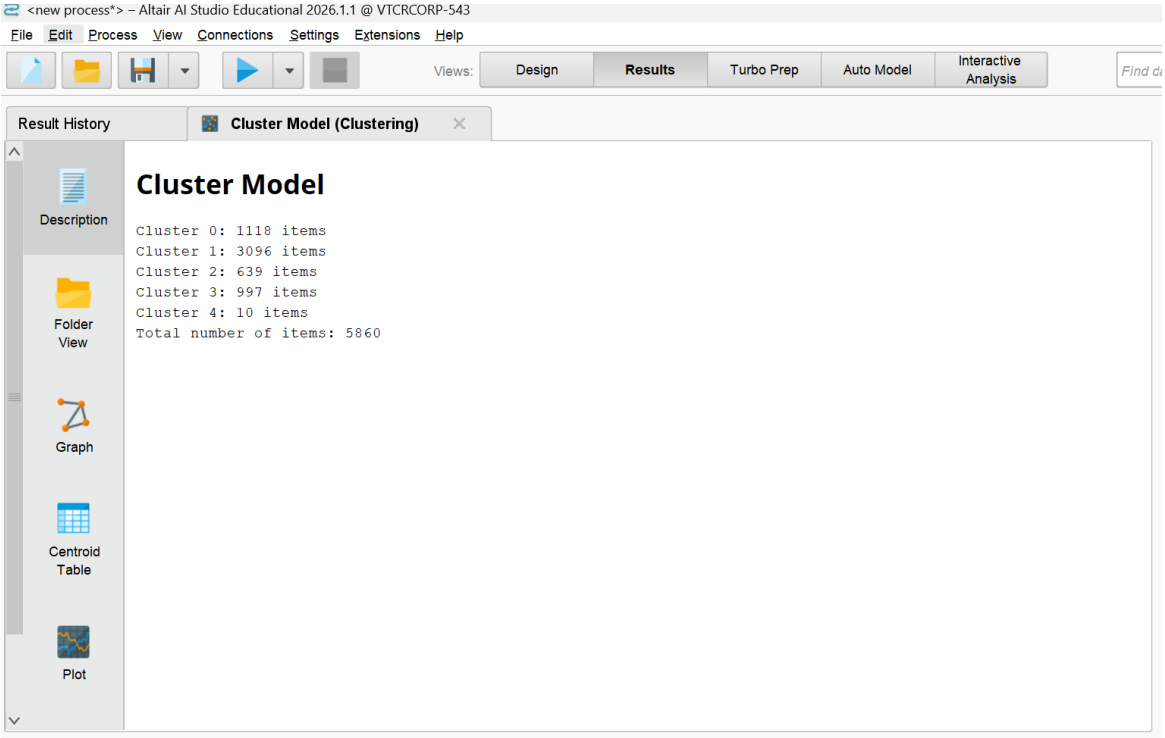


Figure 2. Final Cluster Model output — cluster sizes and total customer count.

Segment	Cluster	Size	% of Total	Recency	Frequency	Monetary
VIP Customers	4	10	0.2%	0.005	0.462	0.361
Loyal Customers	1	3,096	52.8%	0.050	0.015	0.007
Regular Customers	3	997	17.0%	0.286	0.006	0.002

Occasional Customers	0	1,118	19.1%	0.555	0.004	0.002
Inactive Customers	2	639	10.9%	0.841	0.002	0.001

Table 1. Final cluster solution: segment sizes, proportions, and normalized RFM centroids.

Cluster Profiles

The clusters were interpreted by the centroids of the clusters in Recency, Frequency, and Monetary, all scaled to a 0-1 range through Range Transformation. Lower Recency means more recent purchase behavior, and higher Frequency and Monetary mean more frequent and high-value purchases.

VIP Customers (Cluster 4, n = 10)

The highest Frequency (0.462) and Monetary (0.361) values combined with near-zero Recency (0.005), by far. These are the most frequent customers who spend the most and have recently made a purchase on the newest product on the list, the smallest yet the most important customer group.

Loyal Customers (Cluster 1, n = 3,096)

Minimal Recency post-VIP (0.050) with a moderate Frequency and Monetary values. This is the biggest segment so far, which comprises more than half of the customer base and is the mainstay of recurring revenue.

Regular Customers (Cluster 3, n = 997)

Moderate Recency (0.286) with low Frequency and Monetary value; the order of magnitude is similar to that of Occasional and inactive customers but are more active in the recent past.

Occasional Customers (Cluster 0, n = 1,118)

Greater Recency (0.555) than Regular customers, more time delay since the last shopping date, and with equally low Frequency and Monetary values.

Inactive Customers (Cluster 2, n = 639)

The highest Recency (0.841), the most distant time of the last purchase by far, with the least Frequency and Money values. This segment portrays a very low level of current activity.

Visualization of Cluster Differences

The following chart provides a comparison of normalized Recency, Frequency, and Monetary centroids of all five segments, and it is obvious that VIP customers stand out from all the other segments in terms of Frequency and Monetary, whereas Recency is the key factor in dividing the remaining four segments.

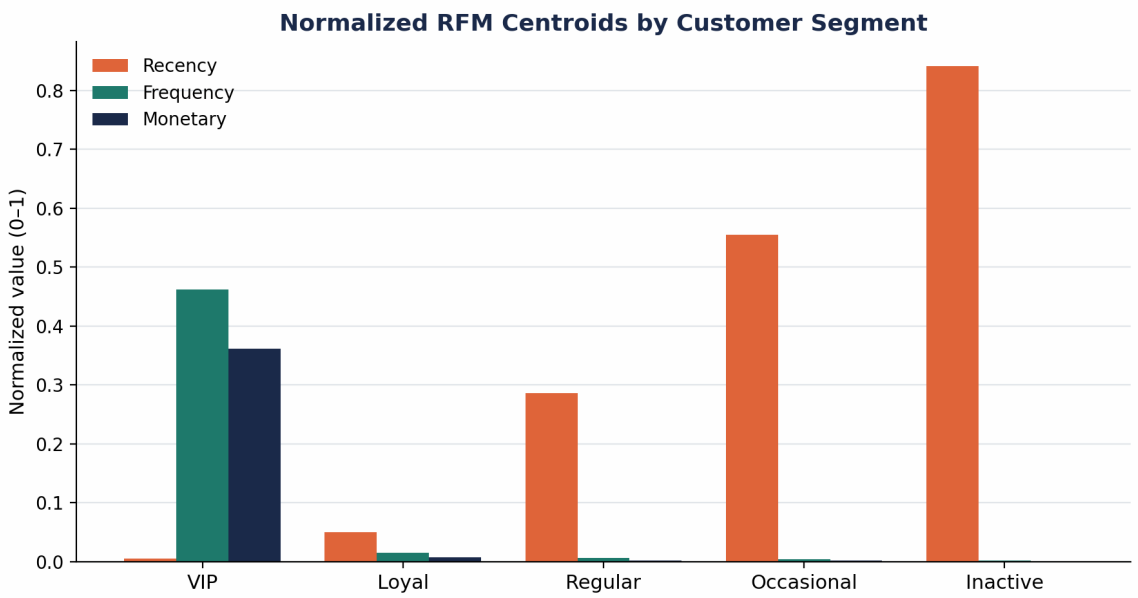


Figure 3. Normalized RFM centroids by customer segment.

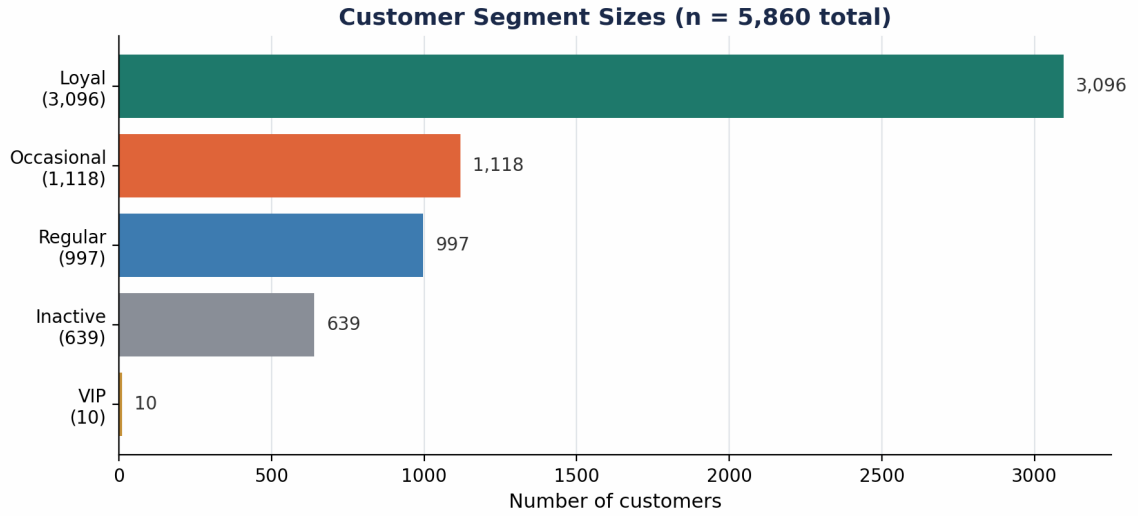


Figure 4. Customer segment sizes (n = 5,860 total).

Scenario Analysis: Comparing Cluster Counts

The group tested the results of K = 3 up to K = 6 to confirm that the final clustering results were accurate and K = 5 was the correct number of clusters, as reported in the midterm phase. The most important trade-off was between K = 4 and K = 5:

K	Observation
3	Clusters too broad — insufficient differentiation between customer types.
4	Better separation than K=3 but high-value customers merged with mid-tier ones — no distinct VIP group.

5 (Final)	Produced five distinct, business-meaningful segments, including an isolated VIP group.
6	Created a very small, low-value cluster with limited business interpretability.

Table 2. Cluster count scenario comparison.

Why K = 3 Was Too Broad

K-Means has only three clusters, resulting in the same customers clustering together, yet they act very differently, merely because there are not enough clusters to cluster them apart. With K = 3, high-value customers (high-value customers) that had very recent, frequent, and high-spending behavior (the VIP profile) were in the same cluster as those with only recent, moderate-frequency, and moderate-spending behavior (the Loyal profile). The resulting cluster centroid will be an average of two actually different types of customers- it does not fairly represent either group.

This has a direct business implication: a marketing intervention targeted at such averaged centroid would be either too costly, granting Loyal customers the VIP treatment, or undersell by supporting the actual VIP customers as being run-of-the-mill repeat purchasers. K = 3 is a mathematically sound choice, but the groups corresponding to it are internally mixed to the level that they cannot be singly actioned according to their distinct needs when in a targeted business action; which is what too broad in this context implies. This was exactly the reason why it was important to raise K to 5: this is where VIPs begin to separate into their own homogeneous group rather than being diluted into a bigger group.

K = 4 makes the model easier but combines the VIP segment into a larger high-value population, and the capability to pinpoint the top ten against the highest-value customers. K = 5 will cut this segment at the expense of one extra cluster to decode. Since the VIP customers (which constitute only 0.2 of the base) have a skewed amount of Monetary value, the team ensured a final business-driven choice of K = 5.

Segment Recommendation Memo

MEMORANDUM

To: Marketing and Customer Relationship Management Team

From: Project 8 Team — Arshiya Ahmed, Ankit Sharma, Bahrom Homidov

Re: Recommended Actions by Customer Segment

Date: [Final Submission Date]

Based on the final K-Means clustering analysis of 5,860 customers, we recommend the following segment-specific actions:

VIP Customers (10 customers, 0.2%)

Give white-glove account management, first-mover benefits, and special perks. This particular group has a small size, but yields disproportionate Monetary value - retaining these should be a high priority.

Loyal Customers (3,096 customers, 52.8%)

Implement a loyalty program. Being the largest segment, any increase in retention or spend here is the aggregate revenue most impactful of any segment.

Regular Customers (997 customers, 17.0%)

Periodic promotions and cross-sell offers can be used to push this portion of the segment to Loyal.

Occasional Customers (1,118 customers, 19.1%)

Now is an opportune moment to launch targeted re-engagement offers, when Recency is mid-range - maintain early intervention costs less than an entire win-back initiative later.

Inactive Customers (639 customers, 10.9%)

Conduct win-back campaigns (offer codes, reminder mail). Prioritise additional customer expenditure on unresponsive customers after one or two attempts to re-allocate the same cash to Occasional and Regular customers.

Business Value of Segmentation

- Minimize marketing investment on the segments that create the most value - Loyal (52.8%) and VIP (0.2%) customers combined, will warrant concentrated retention investment.
- Avoid revenue holes by identifying Occasional customers as they are about to fall into the Inactive segment.
- Individualize marketing at scale — five distinct segments, instead of one-size-fits-all campaigns, individualized and matched to real purchasing behavior.
- Give a repeatable, quantitative foundation (RFM) to the movement of customers between segments over time, not a one-time label.

Limitations

- No statistical measure of cluster validity (e.g., Silhouette Score) was calculated to statistically prove that K = 5 was the best; business interpretability and comparison with scenarios were used to validate the choice.
- RFM variables by themselves are insufficient to understand why customers act the way they do - the model does not contain any product category, channels, or demographic information.
- The fact that the VIP cluster (n = 10) is small might mean that pronounced outliers in the cluster are going to affect the centroid disproportionately.
- Segments are a point-in-time view of the data; the model is not yet concerned with the movement of a single client between segments over time.
- K-Means was not compared with other clustering algorithms (DBSCAN, Hierarchical Clustering) during this stage.

Self-Critique: Are the Clusters Actionable?

The last clusters are viable the way they are. The segments are well partitioned to a clearly defined centroid, and each of them corresponds to a particular varied business action that is recorded in the Segment Recommendation Memo above. The choice of K = 5 (as opposed to the alternative of using K = 4) was made based on an actual trade-off, namely, the isolation of the VIP segment, as opposed to a default assumption.

Meanwhile, a quantitative measure of validity, such as a Silhouette Score, and monitoring segment membership within several periods of time would bolster the belief in the model as being quality and not just a factor of a single time point. These are natural extensions further than what this course project has covered.

Conclusion and Recommendations

During this project, the analysis of raw transaction data through the RFM analysis and Range Transformation normalization was able to transform raw data into five actionable customer prescriptions performed with K-Means clustering in Altair AI Studio (RapidMiner). The last model, which was tested on a K = 4 vs. K = 5 scenario comparison, resulted in business-interpretable segments: VIP, Loyal, Regular, Occasional, and Inactive Customer, with a self-describing distinct recommended action.

Based on the recommendations of the midterm stage, the last recommendations by the team are:

- Introduction: Design marketing campaigns that are specific to each segment and not all customers.
- Protect the segments driving the most value by rewarding VIP and Loyal customers with loyalty programs, exclusive offers, and priority service.
- Head Re-engage Inactive customers with specific win-back exercises before they grow further inactive.
- Upsell Regular and Occasional customers by cross-selling and periodic offers.
- The next step in future work on the topic is to test other clustering algorithms (DBSCAN, Hierarchical Clustering) and formal validity measurements (Silhouette Score) to gain additional confidence in the segmentation.

In general, the project validates that customer segmentation is a viable business application of clustering - a way of refining raw transaction data into a data-based marketing, retention, and resource allocation decision-making framework.

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Appendices

Appendix A: Final Deliverables Checklist

Deliverable	Status
Final RapidMiner Workflow	Completed — see Figure 1
Final Clustered Dataset	Exported separately from Altair AI Studio (Cluster attribute appended to customer-level RFM table)
Final Cluster Profile Table	Completed — see Table 1
Final Visualization Outputs	Completed — see Figures 2–4
Segment Recommendation Memo	Completed — see Segment Recommendation Memo section
Final Report (value, limitations, business actions)	Completed — this document